

Context Selection Is a Partial Order: A Blackwell Theory of Why No Relevance Score Transfers Across Tasks

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Abstract

Choosing what to put in a language model’s prompt (a retrieved passage, a private convention, a repository file, a demonstration) is almost universally posed as a *scalar* ranking problem: score each candidate by relevance, mutual information, $\mathcal{V}$ -usable information, or an information-gain proxy, and take the top- k . We argue this primitive is structurally wrong. Modeling each context source as a statistical *experiment* on the task’s latent requirement, we transport Blackwell’s comparison of experiments [Blackwell, 1953] and Le Cam’s deficiency [Le Cam, 1964, 1986] to context selection. The resulting order over sources is *partial*, and our central claim, an application-level consequence of Blackwell’s theorem, is an *incomparability* result: when two sources are Blackwell-incomparable (neither is a garbling of the other), no scalar score can rank them correctly across all tasks. This gives a formal account of the field’s “relevance-does-not-transfer” folklore. To make the order testable on a real model with a real verifier, we define a decision-rule-restricted deficiency *surrogate*, a value gap evaluated at the fixed decision rule the model induces, and equip it with a distribution-free *uniform certificate* (exact Clopper–Pearson intervals with a Bonferroni correction over the task battery), which we show is the multiplicity correction a naive Dvoretzky–Kiefer–Wolfowitz bound omits. We instantiate the theory on six language models across four vendors (Anthropic, OpenAI, DeepSeek, Moonshot). On all six we certify ($\geq 90\%$ confidence, both directions) that two private context sources are empirically incomparable, and a query-conditioned relevance score mis-ranks them on a confound-controlled trap task. We further report two findings the framework anticipates and the surrogate detects: **(i) anti-monotonicity**, supplying a strict superset of a sufficient source can sharply degrade performance, a certified collapse from 100% to as low as 0% pass rate, which falsifies the monotonicity that scalar context-value scores silently assume; and **(ii) a model-relative value of private context**, $I(S; W \mid \theta)$, that varies continuously across models (the guessability of a fixed convention spans 0%–75% across vendors) and tracks a vendor’s training rather than raw capability. The anti-monotonicity effect is large and robust across the capability ladder but is *not* monotone in capability, and the empirical program rests on a single adversarially constructed source pair under a restricted estimator. The conclusion is methodological: context selection should be treated as a partial-order problem, and a certified deficiency surrogate, not a scalar, is the right operational object. We give the selection rule this implies, PROBE-SELECT, which keeps the certified non-dominated set and diversifies over incomparable coordinates; it provably never retains a dominated source, recovers top- k when sources are totally ordered, and on our measured data avoids the anti-monotonicity trap a superset heuristic falls into. Scaling it beyond curation-size candidate sets to corpus retrieval is the central problem we leave open.

1 Introduction

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information, $\mathcal{V}$ -usable information [Ethayarajh et al., 2022], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. We argue it is also the wrong abstraction, and that a classical apparatus from statistical decision theory says precisely why.

A folklore observation, restated as structure. A persistent, informal observation undercuts the scalar paradigm: *relevance does not transfer across tasks*. A source that helps one task hurts another; “the best context” is task-dependent. This is usually treated as a modeling nuisance to be engineered around with better rankers. We show it is the signature of a *partial order*. We model a context source W as a statistical experiment on the task’s latent requirement S (the object a verifier checks; §3), which lets us import two classical tools. Blackwell’s order ranks experiments by garbling: $W_1 \succeq_B W_2$ iff W_2 ’s signal is a noised (garbled) version of W_1 ’s, and, by the Blackwell–Sherman–Stein theorem, this holds *iff* an idealized decision-maker does at least as well with W_1 as with W_2 on *every* decision problem [Blackwell, 1953, Torgersen, 1991]. The order is *partial*: many source pairs are incomparable, neither garbling the other. Incomparability is exactly the regime in which no scalar summary can be faithful, because a scalar is a total order and a total order extending a strict partial order must invert at least one incomparable pair on some task.

What is and is not new. The mathematical machinery, the Blackwell garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam’s deficiency, is classical and we claim none of it as new. Our contribution is an *application*: transporting comparison-of-experiments to LLM context selection; an incomparability *framing* that turns cross-task non-transfer of relevance from folklore into a stated consequence; and an *estimator*, a certified, decision-restricted deficiency surrogate, that makes the partial order measurable on a real model behind a real verifier. One gap the framing must respect, and that we close, concerns the decision rule. Blackwell’s theorem is a statement about the *optimal* (Bayes) use of an experiment, whereas every quantity we measure, the pass rate $\text{PASS}(W, T) = \mathbb{E}[v_T(\delta_M(T, w))]$, is evaluated at a *single, fixed, sub-optimal* decision rule δ_M induced by a particular model M . “The optimal user of W_2 beats the optimal user of W_1 on some task” does not by itself imply “the fixed model does better with W_2 than with W_1 .” We therefore develop the order and the incomparability statement in a δ_M -restricted category (§3, §5), prove a one-way bridge to the idealized Blackwell order, and confine every empirical claim to the restricted object that the experiments actually probe. Likewise, our estimator $\hat{\delta}_{\mathcal{D}}$ performs no minimization over garblings, so it is a *decision-restricted value-deficiency surrogate*, not a Le Cam deficiency; we name it as such and relate the two formally (§6) rather than borrowing the deficiency’s pedigree.

A latent-coordinate model that makes the order non-vacuous. A theory referee will object, correctly, that our sources are *fixed deterministic strings* that do not vary with the task instance, so a kernel $\kappa_W(\cdot \mid S)$ would be a constant, hence a Blackwell-trivial (uninformative) experiment with a trivial σ -algebra. We resolve this by modeling the latent requirement as a *vector of coordinates* $S = (S_1, S_2, \dots)$ and each source as a deterministic experiment observing a *projection* $\kappa_{W_i}(\cdot \mid S) = \delta_{\pi_i(S)}$. Comparison of experiments then reduces to refinement of the

projections’ sub- σ -algebras (Blackwell for deterministic experiments), and incomparability becomes the statement that neither projection refines the other, a property we establish for our concrete pair (§3, §8). Under this construction the garbling argument is rigorous rather than a metaphor about the logical content of a string.

Empirical findings. We instantiate the theory on two private, equally on-topic conventions about the same module: an API call contract W_1 and a wire-encoding invariant W_2 (§8), with a four-cell verifier battery using executable PASS/FAIL judgments (return code 0 is PASS). This imposes a restriction: the battery is a fixed, adversarially constructed opposition, not a sampled task distribution, and the certified incomparability is carried by one API cell and one encoding cell; we report the effective support explicitly and do not call $\mathcal{D}$ a “distribution.” On this instrument we obtain, on *real* model calls with no modeled numbers:

- **Certified empirical incomparability on all six models.** The uniform certificate clears $\geq 90\%$ confidence in *both* directions: the two-sided lower bounds $L_{\mathcal{D}}(W_1, W_2) / L_{\mathcal{D}}(W_2, W_1)$ are Haiku-4.5 $+0.88 / +0.83$ ($n=80$), Sonnet-4.6 $+0.76 / +0.76$ ($n=40$), Opus-4.8 $+0.55 / +0.55$ ($n=20$), GPT-5.5 $+0.76 / +0.76$ ($n=40$), DeepSeek-V4-Pro $+0.347 / +0.715$ ($n=40$), and Kimi-K2.6 $+0.675 / +0.762$ ($n=40$). Incomparability is therefore not a single-lab artifact.
- **A real relevance score mis-ranks the pair.** On a trap task saturated with encoding vocabulary, query-conditioned lexical relevance scores $\varphi(W_1)=0.36 < \varphi(W_2)=0.44$ (ranking $W_2 \succ W_1$), yet realized PASS ranks $W_1 \gg W_2$ (e.g. 96–100% vs. 0%). On the honest tasks relevance agrees with PASS; it fails *only* where utility diverges from topicality, the precise prediction of the incomparability framing. We state plainly that this is shown for bag-of-words lexical relevance, the scalar the simplest practitioners use, and is not yet replicated with a learned reranker.
- **Certified anti-monotonicity.** The superset $W_{1+} = W_2 ++ W_1$, which literally *contains* W_1 ’s signal, sharply degrades the API cells: Sonnet and Kimi collapse $100\% \rightarrow 0\%$ (interference $\Psi_T = -1.00$, certified upper bound ≤ -0.69); GPT-5.5 certifies $\Psi_T \leq -0.60$ on `api_post_ok`. The interference certificate clears on at least one cell for five of six models (Opus reproduces, $\Psi_T = -0.75 / -0.80$, but $n=20$ is too few to certify; we give the power argument in §8). An order control shows the harm does not recover under reordering, so it is not a lost-in-the-middle artifact. One caveat: the effect is large and robust across the capability ladder but is *not* monotone in capability (Sonnet, the middle model, is the starkest; Opus, the strongest, less so), so we claim no capability trend.
- **A model-relative value of private context.** The none-arm baseline on `api_argorder` forms a continuous spectrum, Claude $\sim 0\%$, DeepSeek 2%, Opus 15%, Kimi 45%, GPT-5.5 75%, so the value $I(S; W \mid \theta)$ of the private context (where θ denotes the model together with its pretraining prior over conventions) varies *continuously* across models and tracks a vendor’s training rather than raw capability (DeepSeek is tight like Claude despite a different vendor). This is an empirical instance of the theory’s θ -dependence, and it caps which cells can be certified for which model.

Methodological asymmetry. The Anthropic models run agentially (`claude -p`, up to six turns); the non-Anthropic models run single-shot (completion plus code extraction via an OpenAI-style API). We flag this throughout: vendor is confounded with harness, so every *cross-vendor magnitude* comparison (the value spectrum, the interference ladder) inherits that confound, whereas

the binary within-model claims do not. The single-shot setting is arguably a *cleaner* probe of whether a model uses supplied context, since it removes the agentic-exploration confound; we treat the asymmetry as a limitation to be reported, not explained away.

Contributions.

- (C1) **A decision-independent partial order over context sources** (§4). We instantiate the Blackwell garbling order on context sources and state the for-all-tasks dominance guarantee for the idealized decision-maker via the Blackwell–Sherman–Stein theorem, together with the δ_M -restricted order the experiments probe and a one-way bridging lemma between them. This is strictly stronger than any scalar relevance score, which is task-specific.
- (C2) **An incomparability result for context selection** (§5). We give an application-level consequence of Blackwell’s theorem: Blackwell-incomparable sources cannot be ranked by *any* query-independent scalar $\varphi(W)$ for all tasks, and we separately establish, as an empirically grounded statement, that even a query-*conditioned* relevance score $\varphi(W, T)$ mis-ranks them on a constructed trap. The theorem covers source-level scalars, and the trap covers the query-conditioned case; the two are distinct. This is the first formal account, to our knowledge as of mid-2026, of cross-task non-transfer of relevance.
- (C3) **A measurable, certified deficiency surrogate** (§6). We define $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = \max(0, \sup_{T \in \mathcal{D}} [\text{PASS}(W_2, T) - \text{PASS}(W_1, T)])$, a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform certificate: exact Clopper–Pearson intervals [Clopper and Pearson, 1934] at Bonferroni level [Bonferroni, 1936] $\alpha = \eta/(2|\mathcal{D}|)$, valid jointly on a single event, so that $L_{\mathcal{D}} > 0$ certifies a positive value gap at confidence $\geq 1 - \eta$. We argue why a naive DKW bound [Dvoretzky et al., 1956] under-covers a supremum over $|\mathcal{D}|$ cells, and we give the matching interference certificate that the anti-monotonicity finding requires.
- (C4) **A constructive partial-order selector** (§7). PROBE-SELECT keeps the certified non-dominated set and then diversifies by certified task coverage. We prove it (i) never retains a Blackwell-dominated source (monotone safety), (ii) recovers scalar top- k when the sources are totally ordered, and (iii) otherwise covers distinct latent coordinates with the standard $(1 - 1/e)$ submodular guarantee. On our measured data it returns the $\{W_1, W_{1+}\}$ front and rejects the superset-only choice that the anti-monotonicity result shows would collapse, answering “what replaces the scalar,” with an explicit account of its curation-scale cost and the corpus-scale open problem.
- (C5) **A cross-vendor empirical program** (§8). Certified incomparability and the relevance mis-rank on six models across four vendors; certified anti-monotonicity on five of six; and a continuous map of the model-relative value of private context, all on real model calls, with the scope limits (single source pair, vendor–harness confound, capability-non-monotonicity) stated in §10.

Roadmap. §2 positions the work against scalar context-selection criteria and the recent transport of comparison-of-experiments into machine learning, and frames our delta as application-plus-framing-plus-estimator rather than new core mathematics. §3 gives the latent-coordinate experiment model, defines θ and the δ_M -restricted order, and states the non-refinement (hence incomparability) construction. §4 develops the partial order (C1) and the bridging lemma. §5 proves the source-level

incomparability statement and isolates the query-conditioned case (C2). §6 defines the surrogate, the uniform certificate, and the interference certificate, and relates the surrogate to the true Le Cam deficiency (C3). §7 gives the PROBE-SELECT selector and its monotone-safety, top- k -generalization, and coverage guarantees, with a worked illustration on the measured data (C4). §8 reports the six-model program with full per-cell tables (C5). §10 states threats to validity and scope, and §11 concludes. Proofs accompany the statements in-line; the bibliography is provisional and every non-classical reference is marked [TO VERIFY] pending a check against arXiv/DBLP (see the author note preceding the references).

2 Related Work

Bibliography provenance (author note). Only three references in this paper have been verified by the authors against their primary sources: Blackwell’s comparison of experiments [Blackwell, 1953], Le Cam’s deficiency and the comparison of statistical experiments [Le Cam, 1964, 1986, Torgersen, 1991], and $\mathcal{V}$ -usable information [Ethayarajh et al., 2022]. Every other reference cited below was surfaced by an automated literature search and is marked [TO VERIFY]; the corresponding .bib entries carry a % TO VERIFY comment. Our novelty claims and our characterization of prior art *must* be re-checked against arXiv/DBLP before submission, including a fresh search for any post-2025 work porting statistical deficiency to language-model context. We flag this explicitly because our own due-diligence pipeline previously over-read a candidate prior-art paper as containing results it does not contain (see the discussion of [Akdemir, 2025] below); the lesson is that machine-summarized prior art is unreliable and a manual full-text read is mandatory.

Positioning. The machinery we use, Blackwell’s garbling order [Blackwell, 1953] and Le Cam’s deficiency [Le Cam, 1964, 1986], is classical, and we claim no new core mathematics. Our contribution is an *application*, a *framing*, and an *estimator*: (i) we treat each candidate context source for a language model as a statistical experiment on the task’s latent requirement and rank sources by the Blackwell order, with a decision-independent, for-all-tasks dominance guarantee (§4); (ii) we observe, as an application-level corollary of Blackwell’s theorem, that Blackwell-*incomparable* sources cannot be ranked by any scalar score for all tasks (§5), which gives the first formal account of the empirical folklore that “relevance does not transfer across tasks”; and (iii) we supply a decision-restricted Le Cam deficiency *surrogate* with a distribution-free finite-sample certificate that is auditable on a real model with a real verifier (§6). To our knowledge, and subject to the caveat above, no prior work applies the comparison of experiments to selecting inference-time context for language models.

2.1 Scalar context and demonstration selection

Information-theoretic context and demonstration selection for language models is dominated by *scalar* criteria, each of which induces a *total* order over candidate sources and takes a top- k prefix. The dominant family scores topical relevance, either lexically or with learned rankers trained jointly with the generator [Yu et al., 2024]. A second family scores *usable* information: $\mathcal{V}$ -usable (predictive- $\mathcal{V}$) information measures how much a fixed model family can extract from an input [Ethayarajh et al., 2022], and recent work uses it to enhance contextual faithfulness in language models [Yuan et al., 2025]. A third family scores expected information or information gain, mutual information and Bayesian experimental design for language-model querying [Choudhury et al., 2025], and information-pursuit variants with conformal stopping [Chan et al., 2025]. A fourth family attributes value to demonstrations through game-theoretic or influence-based valuations [Xie et al., 2024,

Vinay et al., 2024]. A fifth treats the question as binary classification: a “sufficient-context” detector predicts whether the supplied context is enough to answer [Joren et al., 2024].

Every criterion in this family collapses the comparison of two sources to a comparison of two real numbers, and is therefore a *scalar* in the precise sense of Theorem 2: it imposes one total order on $\{W_1, W_2\}$. Our delta is structural, not a better scalar. We show (§5) that when two sources are Blackwell-incomparable, as our two private conventions are (§8), *any* such scalar, including relevance, mutual information, $\mathcal{V}$ -usable information, perplexity-gap, and expected information gain, must mis-rank them on some task. These criteria are not competitors to be out-performed but instances of the class our incomparability result rules out: each is Blackwell-monotone, so each is a one-dimensional projection of a partial order it cannot recover. We substantiate the monotonicity claim for the closest neighbor, $\mathcal{V}$ -usable information, in Proposition 1 rather than asserting it; we make the weaker conjecture that the same holds for the remaining members of the family. Two points sharpen the separation from the strongest neighbors. First, against expected information gain and Bayesian experimental design [Choudhury et al., 2025, Chan et al., 2025], the relevant axis is *decision-dependence*: those criteria are scalar scores optimized for a particular query or stopping rule, whereas our Blackwell guarantee (Theorem 1) is *decision-independent*, it holds for every prior, every bounded utility, and every task simultaneously. Second, against query-conditioned relevance $\varphi(W, T)$, which is not a fixed total order and is therefore *not* ruled out by Theorem 2, we make a separate, empirical claim: a relevance-trap task on which lexical relevance ranks $\varphi(W_2) > \varphi(W_1)$ while realized PASS ranks $W_1 \gg W_2$ (§8). We are careful not to present the trap as covered by the theorem; the theorem governs query-independent scalars, and the trap is the empirical patch for the query-conditioned case.

2.2 Task-dependence of context utility

A parallel literature establishes that the utility of context is task-dependent rather than an intrinsic property of the source, but does so only empirically or informally: some work argues directly that topical relevance is not the same as answer utility [LeVine and Varjavand, 2025, Qiu et al., 2026], and the broader phenomenon that adding more or longer context can *degrade* performance has been documented from several angles [Du et al., 2025, Liu, 2026]. A separate strand observes Pareto-incomparability of prompts, but along competing *metrics within a single task* [Wen et al., 2025], which is orthogonal to our object: incomparable *sources across tasks*. The folklore that the right context is task-dependent, that relevance does not transfer across tasks, is exactly what our contribution converts into a theorem. Cross-task non-transfer is not a modeling nuisance to be engineered around but a necessary consequence of the Blackwell order being partial: any scalar imposes a total order that the true partial order must violate somewhere. The empirical degradation results are likewise reframed: our anti-monotonicity finding (§8) is not merely that “more context can hurt”, an observation already in the literature [Du et al., 2025, Liu, 2026], but that a strict *superset* of a sufficient source can hurt, on sources we have separately *certified* to be Blackwell-incomparable, with a per-task interference certificate; the algebraic object $\Psi_T < 0$ is one that no monotone-submodular context-value scalar can carry.

2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has recently entered machine learning, but exclusively for ranking *tasks* or *domains*, never inference-time context for language models. Statistical deficiency has been used to estimate task inclusion at training time, reconstructing a classic NLP pipeline ordering over *tasks* [Fosse et al., 2025]; this orders the objects a model is trained on, not the context supplied at

inference, and provides neither an incomparability result nor a PASS-rate certificate. A transfer-learning line builds a deficiency-induced *directional* dominance with a finite-sample proxy [Akdemir, 2025]; we read this work in full and confirm it contains no incomparability statement, no formal partial order (only a directional distance), and no distribution-free certificate (it uses an MMD proxy that its own remarks disclaim as a guarantee), and that its domains are vision, genomics, and reinforcement-learning control with no language-model, prompt, or retrieval content. We therefore cite it as the deficiency-in-ML lineage for our order (§4), and differentiate on three axes: object (context sources vs. domains), structure (incomparability vs. directional dominance only), and a distribution-free finite-sample certificate (which the transfer-learning deficiency work does not provide). Comparison-of-experiments ideas have also been connected to loss design and learning from corrupted labels [van Rooyen and Williamson, 2014]; that line concerns the geometry of losses, not representation or context selection, and we cite it only as evidence that the apparatus has a foothold in learning theory.

2.4 The Blackwell order in artificial intelligence

Blackwell’s theorems have recently been surveyed in an artificial-intelligence context [Paxton, 2026], evidence that the comparison-of-experiments apparatus is recognized as relevant to the field; but to our knowledge no prior work applies the Blackwell order to a language model’s *context sources*. We order external context sources supplied to the model, prove the incomparability consequence, and carry out a certified empirical Blackwell-incomparability test across six models and four vendors. To our knowledge no prior work states a Blackwell partial order over context sources, derives the no-scalar-ranks consequence for language models, or supplies a decision-restricted deficiency estimator with a distribution-free certificate, the three pieces this paper contributes.

2.5 Mathematical foundations

We rely on the standard foundations of the comparison of statistical experiments. The equivalence between the garbling order and uniform dominance of Bayes risk is the Blackwell–Sherman–Stein theorem [Blackwell, 1953, Torgersen, 1991]; deficiency and the metric structure of the experiment ordering are due to Le Cam [Le Cam, 1964, 1986], with Torgersen’s monograph as the canonical reference [Torgersen, 1991]. The fact that the Blackwell order is a non-lattice partial order, so that genuinely incomparable pairs exist and no scalar can be order-preserving, is also classical and is documented in the same comparison-of-experiments literature [Torgersen, 1991]. Our incomparability theorem (Theorem 2) is, mathematically, a short corollary of Blackwell’s theorem applied to a constructed pair of decision problems; we state it as an application-level result about language-model context, not as new structure in the order itself. The genuinely new contributions are the framing of context selection as a partial-order problem, the decision-restricted PASS-rate deficiency surrogate, and its distribution-free uniform certificate, none of which the neighbors above provide.

3 Formalization

We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task’s latent requirement. This section fixes the objects and, crucially, separates two decision rules that the informal literature (and an earlier draft of this work) silently conflate: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which is the object every empirical quantity in this paper actually measures. Keeping these distinct is what makes the bridge from theorem to experiment rigorous.

3.1 Tasks, latent requirements, and context sources

A *task* T fixes a latent requirement S_T , the object a deterministic verifier v_T checks (e.g. the intended program behaviour). We take S_T to be a *vector of coordinates*

$$S = (S_1, S_2, \dots, S_m) \in \mathcal{S} = \mathcal{S}_1 \times \dots \times \mathcal{S}_m, \quad (1)$$

where each coordinate S_j encodes one logically independent requirement (in our instantiation, S_1 is the private call-contract convention and S_2 is the private wire-encoding convention; see §8). Modelling S as a product of coordinates, rather than an opaque atom, is the technical device that makes the experiment model well posed for fixed-string sources; we return to this in §3.2 and Lemma 1.

Definition 1 (Context source as an experiment). A *context source* W is a Markov kernel $\kappa_W : \mathcal{S} \rightarrow \Delta(\mathcal{W})$, i.e. a statistical experiment indexed by the latent parameter S , emitting a signal $w \sim \kappa_W(\cdot | S)$ in a signal alphabet $\mathcal{W}$ (the source’s text). The experiment is identified with its kernel; the sample space is $\mathcal{W}$ and the observed datum is the supplied text w .

Definition 2 (Decision rules and PASS). A *decision rule* is a (possibly randomized) map $\delta : (T, w) \mapsto \hat{S}$ producing a candidate solution. Utility is the binary verifier $u(\hat{S}, S) = v_T(\hat{S}) \in \{0, 1\}$. Two decision rules matter:

- (i) the *model rule* δ_M realized by a fixed language model M at a fixed decoding configuration θ (model weights and sampling parameters; see §3.3). Its realized pass rate on task T under source W is

$$\text{PASS}_M(W, T) := \mathbb{E}[v_T(\delta_M(T, w))], \quad w \sim \kappa_W(\cdot | S_T); \quad (2)$$

- (ii) the *Bayes rule* δ_T^* , the (utility-optimal) decision rule for task T given a prior over S , achieving the optimal value $V^*(W, T) := \sup_{\delta} \mathbb{E}[v_T(\delta(T, w))]$.

The distinction in Definition 2 is load-bearing. Blackwell’s theorem (§4) is a statement about V^* (the optimal user of an experiment); every number we report, PASS_M , the deficiency estimator of §6, the interference statistic of §6.3, and the entire six-model program of §8, is a statement about the single fixed rule δ_M . We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*: a snippet stating the private call contract (W_1), or the private wire-encoding invariant (W_2). A naive reading treats $\kappa_W(\cdot | S)$ as a constant kernel δ_{w_0} independent of S . That reading is degenerate: a constant kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial σ -algebra, which would make any garbling argument vacuous. We avoid this by modelling each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

Definition 3 (Projection sources). Fix a partition of coordinate indices. A source W that reveals the coordinates indexed by a set $A \subseteq \{1, \dots, m\}$ is the deterministic kernel

$$\kappa_W(\cdot | S) = \delta_{\pi_A(S)}, \quad \pi_A(S) := (S_j)_{j \in A}, \quad (3)$$

i.e. supplying W lets a decision-maker condition on $\pi_A(S)$ and on nothing outside A . Write $\mathcal{F}_W := \sigma(\pi_A)$ for the sub- σ -algebra of $\mathcal{S}$ generated by the projection. In our instantiation W_1 reveals coordinate S_1 (the call contract, $A_1 = \{1\}$), W_2 reveals S_2 (the wire invariant, $A_2 = \{2\}$), and the superset $W_{1+} = W_2 ++ W_1$ reveals both ($A_{1+} = \{1, 2\}$).

Under Definition 3 a fixed string is *not* an uninformative constant: it is a deterministic experiment whose informativeness is exactly the resolving power of its σ -algebra $\mathcal{F}_W$. The signal alphabet is the range of the projection, and the “text varies with the task” precisely because the realized coordinate values $\pi_A(S_T)$ differ across task instances T . This is the well-posed reading that makes the comparison of §4 a comparison of experiments rather than a metaphor; it is the construction the internal due-diligence record flagged as required and that the earlier draft did not state [Torgersen, 1991].

3.3 The conditioning parameter θ

The model rule δ_M carries an implicit parameter θ : the model’s weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write $I(S; W \mid \theta)$ for the model-relative value of a private source, the information W supplies *beyond what θ already encodes*. When θ already “knows” a convention (the model can guess it without the source), W ’s marginal value collapses; when the convention is private to θ , W is maximally valuable. This θ -dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in §8, where the guessability of a fixed convention spans 0% to 75% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by θ ; we suppress it except where the model-relativity is the point.

4 The Blackwell order over context sources

4.1 The order and its two registers

Definition 4 (Garbling order). $W_1 \succeq_B W_2$ (W_1 *Blackwell-dominates* W_2) iff there is a Markov kernel (garbling) $G : \mathcal{W}_1 \rightarrow \Delta(\mathcal{W}_2)$ with $\kappa_{W_2} = G \circ \kappa_{W_1}$; i.e. W_2 ’s signal is a noised, post-processed version of W_1 ’s, computable from W_1 ’s signal alone without further access to S .

The classical content of the order is the Blackwell–Sherman–Stein theorem, which we state with hypotheses for a self-contained treatment.

Theorem 1 (Blackwell–Sherman–Stein; universal dominance for the optimal rule). *Let W_1, W_2 be experiments on a common parameter space $\mathcal{S}$ with standard Borel signal spaces. The following are equivalent:*

- (i) $W_1 \succeq_B W_2$ (Definition 4);
- (ii) *for every prior μ on $\mathcal{S}$, every finite decision space, and every bounded loss ℓ , the Bayes risk under W_1 is no larger than under W_2 ; equivalently the optimal value satisfies $V^*(W_1, T) \geq V^*(W_2, T)$ for every task T with bounded utility.*

Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii).

Proof. This is Blackwell’s theorem [Blackwell, 1953]; the loss-function form and the role of randomization are standard [Le Cam, 1986, Torgersen, 1991]. We use it as a black box. The only substantive specialization is the reading of “decision problem” as “task with a bounded verifier utility” and “experiment” as “context source” (Definition 1), which is faithful because $v_T \in \{0, 1\}$ is a bounded utility and the kernels of Definition 3 are standard Borel. $\square$

Corollary 1 (Universal context dominance, C1). *If $W_1 \succeq_B W_2$ then, for the optimal user, supplying W_1 is at least as good as supplying W_2 on every task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information, $\mathcal{V}$ -usable information) provides.*

Remark 1 (The order is partial; no lattice). $\succeq_B$ is a genuine partial order: there exist pairs W_1, W_2 with neither $W_1 \succeq_B W_2$ nor $W_2 \succeq_B W_1$, and the order does not form a lattice [Torgersen, 1991]. Incomparability is exactly the regime our experiments target.

4.2 Deterministic sources: incomparability via σ -algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated σ -algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

Lemma 1 (Refinement characterization for projection sources). *Let W_A, W_B be projection sources revealing coordinate sets A, B (Definition 3), with coordinates $S_1, \dots, S_m$ mutually independent under some full-support prior and each S_j non-degenerate. Then:*

- (a) $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$;
- (b) *consequently W_A and W_B are Blackwell-incomparable iff $A \not\subseteq B$ and $B \not\subseteq A$, i.e. each reveals at least one coordinate the other hides.*

Proof. (a) For deterministic experiments a garbling G with $\kappa_{W_B} = G \circ \kappa_{W_A}$ must compute $\pi_B(S)$ from $\pi_A(S)$ alone (no further access to S). Such a measurable map exists iff π_B is $\sigma(\pi_A)$ -measurable, i.e. $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$. Under coordinate independence and non-degeneracy, $\sigma(\pi_B) \subseteq \sigma(\pi_A)$ holds iff every coordinate in B is among those in A , i.e. $B \subseteq A$: if some $j \in B \setminus A$, then S_j is independent of $\pi_A(S)$ and non-constant, so it is not $\sigma(\pi_A)$ -measurable, and no garbling can reconstruct it; conversely if $B \subseteq A$ the coordinate-deletion map is a (deterministic) garbling. (b) is the contrapositive of (a) applied in both directions. $\square$

Lemma 1 is the formal version of the garbling argument used in §8: from the wire format (W_2 , revealing S_2) one cannot manufacture the private arg-order / required-keyword / exception-type facts (S_1), and from the call contract (W_1 , revealing S_1) one cannot manufacture the cents-scale / check-digit facts (S_2). With $A_1 = \{1\}$, $A_2 = \{2\}$ neither set contains the other, so W_1, W_2 are Blackwell-incomparable by construction (and W_{1+} , revealing $\{1, 2\}$, dominates both). This guarantees the *theoretical* incomparability the experiments then exhibit empirically.

4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns V^* . Our experiments measure PASS_M for a single fixed, generally sub-optimal rule δ_M . The two are not interchangeable: an inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a δ_M -restricted order and state precisely what it does and does not inherit from Theorem 1.

Definition 5 (δ_M -restricted, $\mathcal{D}$ -restricted order). Fix the model rule δ_M and a finite task set $\mathcal{D}$. Say $W_1 \succeq_{M, \mathcal{D}} W_2$ iff $\text{PASS}_M(W_1, T) \geq \text{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$. Call W_1, W_2 ($M, \mathcal{D}$)-*incomparable* iff $W_1 \not\succeq_{M, \mathcal{D}} W_2$ and $W_2 \not\succeq_{M, \mathcal{D}} W_1$, i.e. each strictly out-performs the other on some task in $\mathcal{D}$.

$\succeq_{M,\mathcal{D}}$ is the order our certificate measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite $\mathcal{D}$); within those restrictions it is still a genuine, *for-all-tasks-in- $\mathcal{D}$* dominance statement, not a scalar.

Lemma 2 (One-way bridge). *$(M, \mathcal{D})$ -incomparability is strictly weaker than Blackwell incomparability and does not follow from it without a value-attainment hypothesis:*

- (a) *If W_1, W_2 are $(M, \mathcal{D})$ -incomparable for some δ_M and $\mathcal{D}$, then they are Blackwell-incomparable. (Empirical incomparability implies structural incomparability.)*
- (b) *The converse fails in general: Blackwell incomparability concerns V^* , and a fixed sub-optimal δ_M may fail to realize the optimal-value crossover on $\mathcal{D}$. It holds under the value-attainment hypothesis (Assumption 1).*

Assumption 1 (Value attainment on the witnessing tasks). There exist tasks T_a, T_b and a margin $\gamma > 0$ such that δ_M attains $\text{PASS}_M(W_1, T_a) - \text{PASS}_M(W_2, T_a) \geq \gamma$ and $\text{PASS}_M(W_2, T_b) - \text{PASS}_M(W_1, T_b) \geq \gamma$; i.e. the fixed model actually exploits each source’s private coordinate on the task that requires it. Under Assumption 1 with $\{T_a, T_b\} \subseteq \mathcal{D}$, W_1 and W_2 are $(M, \mathcal{D})$ -incomparable.

Proof of Lemma 2. (a) $(M, \mathcal{D})$ -incomparability exhibits a task T_b where $\text{PASS}_M(W_2, T_b) > \text{PASS}_M(W_1, T_b)$. Were $W_1 \succeq_B W_2$, then W_2 ’s signal is a garbling of W_1 ’s, so the rule “garble W_1 ’s signal to W_2 ’s, then apply δ_M ” is available under W_1 and achieves $\text{PASS}_M(W_2, T_b)$ using W_1 ; hence W_1 admits a rule matching W_2 on T_b , while δ_M itself does strictly worse, no contradiction with $W_1 \succeq_B W_2$ by itself, but the symmetric task forbids the reverse garbling. Concretely, a task T_a with the reverse strict inequality $\text{PASS}_M(W_1, T_a) > \text{PASS}_M(W_2, T_a)$ forbids $W_2 \succeq_B W_1$ by the same construction with roles exchanged. Since $(M, \mathcal{D})$ -incomparability supplies both T_a and T_b , neither garbling direction can hold, so W_1, W_2 are Blackwell-incomparable.¹ (b) Assumption 1 directly yields $\text{PASS}_M(W_1, T_a) > \text{PASS}_M(W_2, T_a)$ and the reverse on T_b , which is Definition 5’s incomparability; the failure of the unconditional converse is the content of §6.4 (anti-monotonicity: a fixed rule violating an optimal-value inequality). $\square$

Lemma 2 is the keystone fix. It says the empirical program is sound in the direction we actually need: *a certified $(M, \mathcal{D})$ -incomparability (which our certificate delivers) implies genuine Blackwell incomparability* (part (a)), so the structural claim is earned, not assumed; and it makes explicit that the reverse implication, predicting the realized crossover from the abstract theorem, requires Assumption 1, which our task design enforces (each cell is constructed so the decisive private coordinate is necessary for PASS) and our measurements verify.

5 The incomparability theorem

We now establish that no scalar can rank Blackwell-incomparable sources. We *split* the claim into a clean theorem for query-*independent* source scores $\varphi(W)$ and a separate, explicitly empirical statement for query-*conditioned* scores $\varphi(W, T)$ (the relevance score practitioners actually use); an earlier draft conflated the two and over-claimed.

¹The argument uses only that a garbling is post-processing of the signal, available to *any* fixed rule; it does not require optimality of δ_M . For the deterministic projection sources of Definition 3 the garbled rule is admissible without measure-theoretic subtlety, since the garbling is a measurable coordinate map.

5.1 Query-independent scores: the theorem

Theorem 2 (No query-independent scalar ranks incomparable sources, C2). *Let W_1, W_2 be Blackwell-incomparable. Let $\varphi : \{\text{sources}\} \rightarrow \mathbb{R}$ be any query-independent scalar score (a single number per source: relevance to a fixed corpus, mutual information $I(S; W)$, $\mathcal{V}$ -usable information, perplexity-gap, expected information gain under a fixed prior). Then there exist tasks T_a, T_b such that the ranking of $\{W_1, W_2\}$ induced by φ disagrees with the optimal-value ranking on at least one of T_a, T_b ; and, under Assumption 1, with the realized-PASS_M ranking on at least one of them.*

Proof. By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, “not $W_1 \succeq_B W_2$ ” yields a task T_b (a prior plus bounded utility) with $V^*(W_2, T_b) > V^*(W_1, T_b)$, and “not $W_2 \succeq_B W_1$ ” yields a task T_a with $V^*(W_1, T_a) > V^*(W_2, T_a)$; both inequalities are *strict*, so ties do not arise. A query-independent φ assigns the single ordered pair $(\varphi(W_1), \varphi(W_2))$, fixing one total order of $\{W_1, W_2\}$ *independent of the task*. That fixed order agrees with the optimal-value order on at most one of T_a, T_b (they point opposite ways), hence is violated on the other. The PASS_M conclusion follows by replacing V^* with PASS_M via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin γ . $\square$

Remark 2 (Scope). Theorem 2 rules out exactly the class of *source-level* scalars: any φ that is a fixed number per source, regardless of task. $\mathcal{V}$ -usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1). The theorem is mathematically a one-line consequence of Blackwell’s theorem; the contribution is the *application and framing*, reading cross-task non-transfer of relevance as the signature of a partial order, not new core mathematics [Blackwell, 1953, Torgersen, 1991].

Proposition 1 ($\mathcal{V}$ -usable information is Blackwell-monotone). *Let $\mathcal{V}$ -usable information $I_{\mathcal{V}}(W \rightarrow S)$ be defined, after Ethayarajh et al. [2022], as the reduction in expected predictive Bayes risk of the best predictor in a fixed family $\mathcal{V}$ when given the signal w versus the null signal. If $W_1 \succeq_B W_2$ and the predictor family $\mathcal{V}$ is closed under post-composition with measurable garblings, then $I_{\mathcal{V}}(W_1 \rightarrow S) \geq I_{\mathcal{V}}(W_2 \rightarrow S)$. Hence $I_{\mathcal{V}}$ is a Blackwell-monotone scalar: it respects $\succeq_B$ but collapses it to one real number.*

Proof. Let G be the garbling with $\kappa_{W_2} = G \circ \kappa_{W_1}$. Any predictor $g \in \mathcal{V}$ acting on W_2 ’s signal is realized on W_1 ’s signal by the composite $g \circ G$, which lies in $\mathcal{V}$ by the closure hypothesis. Therefore the best $\mathcal{V}$ -predictor under W_1 attains at least the predictive accuracy of the best under W_2 , so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under W_1 . The scalarity is immediate: $I_{\mathcal{V}}$ returns a single real number per source, so it induces a total order and, by Theorem 2, cannot rank a Blackwell-incomparable pair correctly for all tasks. $\square$

Remark 3 (Scope of Proposition 1). The closure hypothesis (a family closed under post-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour, $\mathcal{V}$ -usable information, and conjecture, rather than prove, the analogue for mutual information, perplexity-gap, and fixed-prior EIG, each of which is a data-processing-monotone functional of the same kernels.

5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score $\varphi(W, T)$, e.g. the lexical or embedding relevance of a source to the task prompt, which is what retrieval and RAG actually deploy, induces a *different* order per task and

is therefore *not* a single fixed total order. Theorem 2 does *not* cover it. We make two precise statements instead.

Proposition 2 (Constrained impossibility for feature-measurable scores). *Suppose $\varphi(W, T)$ is measurable with respect to a fixed feature map of the pair (W, T) that is topical: it depends on W, T only through surface lexical overlap and is invariant to the private coordinate $\pi_A(S_T)$ that the verifier checks. If there is a task T_c whose decisive coordinate is supplied by W_1 but whose surface lexicon is saturated with W_2 ’s vocabulary, then $\varphi(W_2, T_c) > \varphi(W_1, T_c)$ while the realized utility ranks $W_1 \succ W_2$ on T_c . Hence no topical query-conditioned score ranks $\{W_1, W_2\}$ correctly on all tasks.*

Proof. By topical invariance, $\varphi(\cdot, T_c)$ ignores the verifier-decisive coordinate and orders by lexical overlap, which favours W_2 on T_c by construction; utility favours W_1 because T_c ’s PASS requires W_1 ’s coordinate. The two orders disagree on T_c . $\square$

Proposition 2 is the strongest *general* statement we can make about query-conditioned scores: it holds for any score in the topical (lexical-feature) class, but not for an arbitrary learned reranker that could, in principle, read the private coordinate. We therefore treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use. In §8 the `trap_store_wire` task realizes T_c : bag-of-words cosine gives $\varphi(W_1) = 0.36 < \varphi(W_2) = 0.44$ (ranking $W_2 \succ W_1$), yet realized PASS ranks W_1 at 96%–100% versus W_2 at 0% across all six models. The claim is restricted to lexical relevance; we do not claim to have defeated every learned reranker.

6 The decision-restricted deficiency surrogate and its uniform certificate

6.1 The surrogate and its correct name

Exact garblings over text are uncomputable and $\succeq_B$ is partial, so we measure the δ_M -restricted order of Definition 5. Fix the model rule δ_M and a finite task set $\mathcal{D}$ and define the *decision-restricted value-deficiency* (a one-sided worst-case regret of W_1 relative to W_2 on $\mathcal{D}$):

$$\hat{\delta}_{\mathcal{D}}(W_1, W_2) := \max\left(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)]\right), \quad (4)$$

estimated per cell by $\text{PASS}_M(W, T) \approx k_{W,T}/n$ from n completions. Then $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ certifies empirical dominance $W_1 \succeq_{M, \mathcal{D}} W_2$, and mutual positivity $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$ certifies $(M, \mathcal{D})$ -incomparability.

Remark 4 (This is *not* Le Cam deficiency). The Le Cam deficiency $\delta(\mathcal{E}, \mathcal{F}) = \inf_G \sup_{\text{loss}} [R_{\mathcal{F}} - R_{G \circ \mathcal{E}}]$ minimizes over garblings G and takes a supremum over a class of losses [Le Cam, 1964, 1986, Torgersen, 1991]. The quantity in (4) performs no garbling minimization and uses a single fixed rule δ_M over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates $\hat{\delta}_{\mathcal{D}}$ to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (4) the *decision-restricted value-deficiency* and use “Le Cam deficiency” only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative: $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ realizes the for-all-tasks-in- $\mathcal{D}$ dominance of Definition 5, the measurable shadow of Corollary 1.

6.2 The uniform certificate

The surrogate (4) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

Proposition 3 (Uniform incomparability certificate). *Within each cell (W, T) draw n i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let $[L_{W,T}, U_{W,T}]$ be the exact (two-sided) Clopper–Pearson interval [Clopper and Pearson, 1934] for $p_{W,T} = \text{PASS}_M(W, T)$ at level $\alpha = \eta/(2|\mathcal{D}|)$. By the Bonferroni/union bound [Bonferroni, 1936] over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T \in \mathcal{D}}$, all $2|\mathcal{D}|$ intervals hold simultaneously with probability at least $1 - \eta$. On that event,*

$$L_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [L_{W_2,T} - U_{W_1,T}] \quad (5)$$

is a $(1 - \eta)$ lower confidence bound on $\sup_{T \in \mathcal{D}} [p_{W_2,T} - p_{W_1,T}]$. Hence $L_{\mathcal{D}}(W_1, W_2) > 0$ certifies $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ at confidence $\geq 1 - \eta$. Both directions $L_{\mathcal{D}}(W_1, W_2)$ and $L_{\mathcal{D}}(W_2, W_1)$ read off the same $2|\mathcal{D}|$ intervals, so the joint incomparability claim $\{\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0\}$ holds at $\geq 1 - \eta$ with no additional correction.

Proof. Let E be the event that all $2|\mathcal{D}|$ Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most $\alpha = \eta/(2|\mathcal{D}|)$ (exact coverage of Clopper–Pearson [Clopper and Pearson, 1934]), so by the union bound $\Pr[E^c] \leq 2|\mathcal{D}| \cdot \alpha = \eta$, giving $\Pr[E] \geq 1 - \eta$. On E , for every T we have $p_{W_2,T} \geq L_{W_2,T}$ and $p_{W_1,T} \leq U_{W_1,T}$, so $p_{W_2,T} - p_{W_1,T} \geq L_{W_2,T} - U_{W_1,T}$; taking $\max_T$ on both sides, $\sup_T [p_{W_2,T} - p_{W_1,T}] \geq L_{\mathcal{D}}(W_1, W_2)$. Thus on E (probability $\geq 1 - \eta$) the true sup is at least $L_{\mathcal{D}}(W_1, W_2)$, which is the claimed lower confidence bound; if it is positive, the true sup is positive, certifying $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$. The reverse direction uses $L_{W_1,T}$ and $U_{W_2,T}$, which are endpoints of the *same* $2|\mathcal{D}|$ intervals already included in E ; therefore both one-sided certificates are valid on the single event E , and their conjunction holds at $\geq 1 - \eta$ without a further Bonferroni factor. $\square$

Remark 5 (Why a naive DKW bound fails). The Dvoretzky–Kiefer–Wolfowitz inequality [Dvoretzky et al., 1956] bounds the sup-deviation of *one* empirical CDF from *its own* population CDF over a single i.i.d. sample. Our object is different in two ways that DKW does not handle: it is a max over $|\mathcal{D}|$ *differences of two binomial means across distinct cells*, and the max *selects* the *argmax-gap* cell. A single-sample sup inequality neither spans distinct cells nor pays for the selection, so its nominal level under-covers a supremum over $|\mathcal{D}|$ comparisons; the Type-I error grows with $|\mathcal{D}|$. The multiplicity must be paid explicitly via the union bound (Proposition 3), not absorbed into a single-sample DKW envelope. This is the correction over the earlier sketch.

Assumption 2 (Sampling regime). The n outcomes within a cell are i.i.d. Bernoulli draws of $v_T(\delta_M(T, w))$. This holds when completions are sampled independently at fixed temperature/top- p with independent randomness per draw; for the single-shot arms (one completion + deterministic code extraction per draw) the draws are manifestly independent. For the agentic arms (**claude-p**, multi-turn) independence is across *trajectories*: each of the n draws is a fresh trajectory with no shared state, so the binomial model applies to the trajectory-level PASS outcome; we report decoding configuration and defend this independence in §8. Where draws are near-deterministic ($p \in \{0, 1\}$ stark cells), Clopper–Pearson collapses and the certificate is conservative.

6.3 The interference certificate (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source W_{1+} (revealing strictly more coordinates than W_1 ; Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T := \text{PASS}_M(W_{1+}, T) - \text{PASS}_M(W_1, T) - \text{PASS}_M(W_2, T) + \text{PASS}_M(\text{none}, T), \quad (6)$$

the second-difference (interaction) of adding W_2 on top of W_1 . We write the interaction as Ψ_T (rather than a symbol I that would collide with mutual information $I(S; W \mid \theta)$). A certified $\Psi_T < 0$ on a cell whose W_1 arm is at ceiling is the localized statement that “more strictly-more-informative context hurts.”

Proposition 4 (Interference certificate). *Fix a target task $T^* \in \mathcal{D}$ chosen before unblinding. With per-cell exact Clopper–Pearson endpoints $\text{CP}_{\text{lo}}, \text{CP}_{\text{hi}}$ at level $\alpha = \eta/4$ on the four arms $\{W_{1+}, W_1, W_2, \text{none}\}$ of T^* ,*

$$\bar{\Psi}_{T^*} := \text{CP}_{\text{hi}}(p_{W_{1+}}) - \text{CP}_{\text{lo}}(p_{W_1}) - \text{CP}_{\text{lo}}(p_{W_2}) + \text{CP}_{\text{hi}}(p_{\text{none}}) \quad (7)$$

is a $(1 - \eta)$ upper confidence bound on Ψ_{T^} . Hence $\bar{\Psi}_{T^*} \leq -\tau$ certifies harmful non-monotonicity $\Psi_{T^*} \leq -\tau$ at confidence $\geq 1 - \eta$. If instead the harmful cell is selected from $|\mathcal{D}|$ candidates after unblinding, use $\alpha = \eta/(4|\mathcal{D}|)$; this is the same selection correction the certificate applies to the deficiency surrogate (Remark 5), and omitting it would repeat the multiplicity error we criticize.*

Proof. The four arms are independent Bernoulli samples (separate completions), so on the intersection of the four Clopper–Pearson events, probability $\geq 1 - 4\alpha = 1 - \eta$ by the union bound, each true proportion lies in its interval. Substituting the coverage-respecting endpoint for each term of (6) (upper endpoints for the positively-signed $p_{W_{1+}}, p_{\text{none}}$, lower endpoints for the negatively-signed p_{W_1}, p_{W_2}) gives $\Psi_{T^*} \leq \bar{\Psi}_{T^*}$ on that event. For a post-hoc selected cell, replacing α by $\eta/(4|\mathcal{D}|)$ and union-bounding over the $4|\mathcal{D}|$ endpoints preserves $\geq 1 - \eta$ simultaneous coverage. $\square$

Remark 6 (The leaky-baseline limitation, stated as a protocol). The upper bound (7) includes $+\text{CP}_{\text{hi}}(p_{\text{none}})$, so it can certify harm only on cells whose **none**-baseline floors near 0 (the convention is unguessable for the model under test). When the baseline is non-zero, e.g. `api_argorder` at **none** = 75% for GPT-5.5, 45% for Kimi, 15% for Opus, the bound is mechanically slack and harm is not certifiable even when the point estimate is strongly negative. We therefore adopt the explicit protocol: *certify Ψ_T only on cells whose **none**-baseline floors near 0 for the model under test*, and report, per model, which cell carried the certificate and what its baseline was. This makes the cell selection a stated rule rather than an outcome-driven choice. The same model-relative guessability that defines the value of private context (§3.3) thus governs which cells can carry the interference certificate.

6.4 Anti-monotonicity as information non-monotonicity of δ_M

A certified $\Psi_{T^*} < 0$ on a ceiling- W_1 cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

Theorem 3 (Information non-monotonicity of the model rule). *Let W_{1+} reveal a superset of W_1 ’s coordinates, so $W_{1+} \succeq_B W_1$ (Lemma 1). Any Blackwell-coherent rule, one whose value is monotone under $\succeq_B$, as the Bayes rule δ^* is by Theorem 1, satisfies $\text{value}(W_{1+}, T) \geq \text{value}(W_1, T)$ for every*

task T . If a fixed model rule δ_M has a task T^* with $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$, then δ_M is not Blackwell-coherent: it violates monotonicity under the garbling order on T^* . In particular no monotone-submodular scalar context-value score can predict δ_M 's behaviour on T^* .

Proof. $W_{1+} \succeq_B W_1$ by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)). By Theorem 1 any rule whose value respects $\succeq_B$ weakly improves under the dominating experiment. The hypothesized strict decrease $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$ contradicts monotonicity, so δ_M is not Blackwell-coherent. A monotone-submodular scalar ϕ assigns $\phi(W_{1+}) \geq \phi(W_1)$ (monotonicity), hence orders W_{1+} above W_1 and cannot reproduce the realized reversal. $\square$

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent, which is precisely the gap Lemma 2(b) warned cannot be closed without value attainment. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.

6.5 Family-wise error across the certified claims

Each certificate above controls error *marginally* at η for its own claim (incomparability via Proposition 3; interference via Proposition 4). When several claims are read off the *same* n -sample run at the same η , incomparability (both directions, on one shared event E), one-directional dominance, and the interference cells, the conjunction over distinct claims is *not* jointly controlled at η unless the budget is split across claims. We state results as marginally certified per claim and, where a joint statement is wanted, allocate η across the C certified claims (e.g. η/C each) so the global family-wise error is $\leq \eta$. The incomparability pair is the one exception that is genuinely joint at η on a single event, as proved in Proposition 3; all other cross-claim conjunctions on a shared run are reported as marginal.

6.6 Power and the role of n

The asymmetric sample sizes in §8 (Opus at $n = 20$ certifies incomparability but not interference) are principled, not ad hoc. For a stark cell with empirical proportion at an endpoint ($\hat{p} \in \{0, 1\}$), the Clopper–Pearson interval collapses to a one-sided bound that clears 0 at small n : the incomparability certificate (5) is carried by such stark cells (W_1 at $\approx 100\%$, W_2 at 0% and vice versa), so $n = 20$ suffices. The interference certificate (7) mixes a possibly-intermediate **none** baseline and the W_{1+} arm; with non-endpoint proportions the intervals are wide and $n = 20$ is underpowered. The two certificates therefore have genuinely different effective- n requirements, which is why the consolidated record marks Opus incomparability as certified and Opus interference as reproduced but not certified.

7 Deficiency-aware context selection

The partial order is not only a diagnosis; it prescribes a selection rule. If sources form a genuine partial order, the correct primitive is not “score and take the top k ” but “keep the certified non-dominated set, then diversify over what remains.” We give that rule, prove three properties, exhibit it on our measured data without running a new experiment, and state its cost and scope. The point is to convert the impossibility result of §5 and the anti-monotonicity of §6 into a constructive, certificate-backed procedure.

7.1 The certified empirical dominance relation

Fix the model rule δ_M and a probe battery $\mathcal{D}$. Write $W_i \supseteq W_j$ (“ W_i empirically dominates W_j ”) when the certificates of §6 jointly establish, at confidence $\geq 1 - \eta$, that (a) $\hat{\delta}_{\mathcal{D}}(W_i, W_j) = 0$ up to the equivalence margin α_{tol} (no probe task on which W_j beats W_i by more than the margin), and (b) $L_{\mathcal{D}}(W_j, W_i) > 0$ (W_i strictly beats W_j on some probe task). This is exactly the one-directional dominance the dominance control certifies. Sources that are mutually non-dominated are *certified incomparable* (both $L_{\mathcal{D}} > 0$); they form the empirical Pareto front.

7.2 The selector

Algorithm 1 (Probe-Select). *Input:* candidate sources $\mathcal{C}$, probe battery $\mathcal{D}$, budget k , model δ_M , level η .

1. Measure $\text{PASS}(W, T)$ for all $W \in \mathcal{C}, T \in \mathcal{D}$ (n draws per cell); form the $2|\mathcal{C}||\mathcal{D}|$ Clopper–Pearson intervals once.
2. Build $\supseteq$ on $\mathcal{C}$ from those intervals (Propositions of §6); no extra model calls.
3. *Prune* every W that is certified-dominated by some $W' \in \mathcal{C}$.
4. Let P be the surviving certified non-dominated set (the empirical Pareto front).
5. If $|P| \leq k$ return P ; else greedily add to the output the survivor with the largest certified marginal PASS gain on probe tasks not yet covered by the chosen set, breaking ties by smallest $\hat{\delta}_{\mathcal{D}}$ to the chosen set, until k are selected.

7.3 Guarantees

Proposition 5 (Monotone safety). *PROBE-SELECT never returns a source certified-dominated by an available source. Hence, with probability $\geq 1 - \eta$, it never discards a source that strictly wins a probe task in favor of one that is nowhere better than it by more than α_{tol} .*

Proof. Step 3 removes only certified-dominated sources, and step 5 selects from the survivors P . By the dominance certificate (§6), a pruned W has, on the joint $(1 - \eta)$ event, no probe task on which it beats its dominator by more than α_{tol} ; the dominator is retained or itself Pareto-equivalent, so no strict probe winner is sacrificed for a uniformly no-better source. $\square$

Proposition 6 (Strict generalization of top- k). *If $\supseteq$ totally orders $\mathcal{C}$ (every pair certified-comparable), the Pareto front is a chain and PROBE-SELECT returns its top k elements, recovering scalar top- k selection. The procedure therefore extends, rather than competes with, the scalar pipeline: it differs only on the incomparable pairs the scalar provably mis-handles (Theorem 2).*

Proposition 7 (Coverage over incomparable coordinates). *On an incomparable survivor set the greedy step (5) maximizes, up to the standard $(1 - 1/e)$ factor of monotone submodular coverage, the number of distinct probe tasks solved by the selected set, i.e. the number of distinct latent coordinates covered, rather than retaining sources that solve the same tasks. This is the partial-order-grounded analogue of maximal-marginal-relevance diversity, with “diversity” defined by certified task coverage rather than embedding distance.*

7.4 Worked illustration on the measured data

Consider $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ with $W_{1+} = W_2 ++ W_1$. The *idealized* Blackwell order would collapse this set to $\{W_{1+}\}$ alone: W_{1+} literally contains both signals, so $W_{1+} \succeq_B W_2$ and $W_{1+} \succeq_B W_1$ (W_1 is the garbling that drops the W_2 text), and a “more signal is strictly better” heuristic keeps only the superset. On a real fixed model this is exactly wrong: the certified *empirical* relation is different because of the anti-monotonicity of §6. From the measured pass rates (Table 4), W_1 beats W_{1+} on the API cells (e.g. Haiku 98% vs. 24%), so $W_{1+} \not\succeq W_1$; W_{1+} beats W_1 on the encoding cell; and both dominate W_2 . PROBE-SELECT therefore prunes only W_2 and returns the front $\{W_1, W_{1+}\}$, whereas the superset heuristic returns $\{W_{1+}\}$ and collapses on every API task. The selector converts the anti-monotonicity finding into a selection-time safeguard: by ordering on the decision-restricted surrogate $\hat{\delta}_{\mathcal{D}}$ rather than on idealized signal content, it keeps the component source whose value the superset destroys. No new measurement is required for this conclusion; it reads directly off the certified $n=80$ Haiku cells.

7.5 Cost and scope

Building $\succeq$ on $m = |\mathcal{C}|$ candidates costs $m \cdot |\mathcal{D}| \cdot n$ verifier-gated model calls (the pairwise certificates reuse a single per-(source,task) PASS table, so the cost is linear in m , not quadratic), plus $O(m^2)$ interval comparisons. This is *curation scale*: selecting among a repository’s handful of private-convention documents, or curating a few-shot pool of tens to hundreds of exemplars. It is *not* corpus-scale retrieval over 10^6 passages, and we do not claim it is. Two routes to scale, both future work, are immediate: (i) amortize the probe by learning a deficiency predictor $\hat{\delta}_{\varphi}(W_i, W_j)$ from cheap features, converting the online $m \cdot |\mathcal{D}| \cdot n$ cost into offline training plus $O(m^2)$ inference; and (ii) use a cheap scalar retriever to pre-filter a shortlist and run PROBE-SELECT only as a partial-order re-ranking head, where the incomparable-set size is small. We validate the certified estimator that powers steps 1–3 in §8; a corpus-scale evaluation of the full selector against strong reranker baselines is the central open problem we identify in §10, not a result we report.

8 Experiments

We instantiate the theory of §§4–6 on real language models. The design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (§8.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every certified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (§8.4), so the certificate is auditable rather than asserted.

8.1 Design: two incomparable context sources

We define two *private, unguessable* conventions on the *same* module `ledger`, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module, same task shape). Because both are on-topic for any “ledger” task, any lexical, embedding, mutual-information, or $\mathcal{V}$ -usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

W_1 , **API call contract.** `ledger.post(account, amount, *, memo)`: `account` first, `amount` second positional, `memo` a *required keyword-only* argument; returns an integer entry id ≥ 1 ; raises

`ledger.PostError` (not `ValueError/KeyError`) on failure.

W_2 , **wire-encoding invariant**. Amounts serialize to `<sign><minor-unit-digits>#<check>`, where `check = (sum of digits) mod 10`; e.g. $\$12.30 \mapsto "+1230\#6"$ and $-\$0.07 \mapsto "-7\#7"$. Decoding rejects a wrong check digit.

W_{1+} , **literal superset**. $W_{1+} = W_2 ++ W_1$, the concatenation of W_2 's signal with W_1 's. As a string superset of W_2 , it Blackwell-dominates W_2 ($W_{1+} \succeq_B W_2$); it is the anti-monotonicity / dominance probe. We also use $W_{1+}^{\text{rev}} = W_1 ++ W_2$ (W_1 first) as an order control.

Why incomparable (garbling argument). A garbling acts on the *signal alone*. From W_2 's wire format one cannot manufacture the private argument order, the required keyword-only `memo`, or the exception type; from W_1 's contract one cannot manufacture the cents scale, the `#` separator, or the mod-10 check. The two facts are logically independent coordinates of the latent requirement S , so neither signal's σ -algebra refines the other's, neither sufficiency direction holds, and W_1, W_2 are Blackwell-incomparable. This is the concrete instance of the coordinate/projection construction of §3: W_1 and W_2 are deterministic experiments observing distinct, mutually non-refining projections of S (Lemma 1).

Confound defense: the double crossover. The experiment's signature is a *double crossover* measured against a **none** baseline (no convention supplied): W_1 lifts the API tasks but *not* the encoding task, while W_2 lifts the encoding task but *not* the API tasks. If the encoding family were merely "harder," W_1 would not lift it and W_2 would still have to lift its own family from the same floor. Matched **none** baselines that floor near zero on both families therefore establish equal intrinsic hardness, so any divergence is attributable to the matching private fact, not to a difficulty asymmetry.

8.2 Tasks

The full design comprises seven tasks with stub-`ledger` verifiers (return code 0 = PASS): four W_1 -decisive API tasks (`api_post_ok`, `api_argorder`, `api_error_type`, `api_memo_required`), two W_2 -decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

Certified battery. The certified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the certified battery. The four-cell selection was made to retain at least one W_1 -decisive, one W_2 -decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (§8.4.1).

The relevance trap. The trap `trap_store_wire` is the confound-proof form of Theorem 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks $W_2 > W_1$; but its graded requirement is the W_1 call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks $W_1 > W_2$. This defeats the query-conditioned

relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine $\varphi_{\text{lex}}(W, T)$ between a source and the task prompt, computed *before* the model is run. This is the lexical-relevance instance of Proposition 2; the empirical mis-rank is shown for this one concrete scalar (§10), while the theorem (Theorem 2) is the general claim for query-independent scalars.

8.3 Methodology

Models and vendors. We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6.

Harness and arms. The harness (`tokenbench/measure_blackwell.py`) is Python-standard-library only (zero external dependencies); it computes $\hat{\delta}_{\mathcal{D}}$ in both directions, the uniform Clopper–Pearson/Bonferroni certificate, the interference certificate, and the lexical relevance score, and it implements the launchers for the Anthropic CLI, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms $\{\text{none}, W_1, W_2, W_{1+}\}$, with W_{1+}^{rev} added for the order-control run. PASS is measured over n completions per cell.

Agentic vs. single-shot harness (a stated confound). The Anthropic models run *agentially* (`claude -p`, up to six turns with intermediate verifier feedback); the non-Anthropic models run as *single-shot* completion plus code extraction through an OpenAI-style API. Vendor is therefore confounded with harness in this study: every Anthropic model is agentic and every non-Anthropic model is single-shot. We treat this as a genuine limitation of the cross-vendor *magnitude* comparisons (§10); the per-model *binary* claims (incomparability certified, relevance mis-rank, anti-monotonicity certified) are within-model and unaffected. We note that the single-shot setting is arguably the cleaner probe of whether a model uses supplied context, since it removes the agentic-exploration pathway.

Certificate parameters. All certificates use confidence budget $\eta = 0.10$. Incomparability uses exact Clopper–Pearson intervals at per-cell level $\alpha = \eta/(2|\mathcal{D}|)$ over the $2|\mathcal{D}|$ proportions $\{p_{W_1, T}, p_{W_2, T}\}_{T \in \mathcal{D}}$, so by the union bound all hold jointly with probability $\geq 1 - \eta$ and both directions read off the same intervals. The interference certificate uses $\alpha = \eta/4$ and the threshold $\tau = 0.30$, certifying $\Psi_T \leq -\tau$ via the upper bound of (7). We report each certificate as marginally controlled at η ; we do not claim joint family-wise control across the incomparability, dominance, and interference claims drawn from a single run (§6.5, §10). The intervals are degenerate at $p = 0$ and $p = 1$, which is why the stark 100%/0% cells certify incomparability at small n while the interference contrast, which mixes intermediate baselines, does not.

Sampling regime. The certificate models each cell as n Bernoulli PASS draws (Assumption 2); the four arms within a cell are separate completions, hence independent. We flag the exact decoding parameters (temperature, top- p , seeding) and a defense of per-draw independence for the multi-turn agentic arm as a required protocol addition before submission (§10); the binomial model underlying every $L_{\mathcal{D}}$, $U_{\mathcal{D}}$, and Ψ_T bound depends on it.

Spend. The certified record reflects roughly \$170 of Anthropic spend (the live `claude -p` runs) plus Azure quota for the non-Anthropic models; the Azure API returns no cost field, so the \$170 figure should not be read as the total cost across all six models.

Table 1: Consolidated 6-model / 4-vendor verdict. Anthropic rows are agentic (`claude -p`, up to six turns); the others are single-shot (§8.3). “Anti-mono. certified” counts cells on which the interference upper bound clears $-\tau = -0.30$ at $\eta = 0.10$. “Argorder `none`” is the no-context baseline PASS on `api_argorder`, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers.

Claim	Haiku	Sonnet	Opus	GPT-5.5	DeepSeek	Kimi-K2.6
Vendor	Anthropic	Anthropic	Anthropic	OpenAI	DeepSeek	Moonshot
Harness	agentic	agentic	agentic	single-shot	single-shot	single-shot
n per cell	80	40	20	40	40	40
Incomp. certified ($\geq 90\%$)	✓	✓	✓	✓	✓	✓
$L_{\mathcal{D}}(W_1, W_2)$	+0.88	+0.76	+0.55	+0.76	+0.347	+0.675
$L_{\mathcal{D}}(W_2, W_1)$	+0.83	+0.76	+0.55	+0.76	+0.715	+0.762
Relevance mis-rank (trap)	✓	✓	✓	✓	✓	✓
Anti-mono. certified (# cells)	2	2	0 (repro.)	1	1	1
<code>api_argorder none</code>	$\sim 0\%$	0%	15%	75%	2%	45%

8.4 Results

We report four findings: incomparability (§8.4.1), the relevance mis-rank (§8.4.2), anti-monotonicity (§8.4.3), and the model-relative value of private context (§8.4.4). Table 1 consolidates the per-model verdicts; Tables 2, 3, and 4 give the underlying quantities.

8.4.1 Incomparability certified on all six models

On every model the empirical deficiency is mutually positive, $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ and $\hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$, and the uniform lower bounds $L_{\mathcal{D}}$ in both directions clear zero, certifying empirical incomparability at $\geq 90\%$ (Table 2). The crossover is stark: W_1 solves the API and trap cells and fails encoding, while W_2 solves encoding and fails the API and trap cells. For Haiku at $n = 80$ the carrying cells are `api_post_ok` (`none` 0% / W_1 98% / W_2 0%) and `enc_amount` (`none` 0% / W_1 0% / W_2 100%); the certificate strengthened from a marginal $L_{\mathcal{D}} = +0.138$ at $n = 10$ to $L_{\mathcal{D}} = +0.877 / +0.831$ at $n = 80$, far clear of zero.

Because incomparability is a two-directional claim, each row of Table 2 requires both bounds positive; the smaller of the two is the binding one. The two carrying cells are one API cell and one encoding cell, so the effective support of the surrogate is two cells. We therefore describe $\mathcal{D}$ as a fixed adversarial four-cell battery rather than a sampled task distribution: the certificate is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition replicated across six models.

Why $n = 20$ certifies incomparability for Opus but not interference. Opus is run at $n = 20$, which certifies incomparability ($L_{\mathcal{D}} = +0.552$ both directions) but not anti-monotonicity. This asymmetry is principled, not an inconsistent threshold (§6.6). The incomparability bound is carried by stark cells whose proportions sit at 0 or 1, where the Clopper–Pearson interval is degenerate and a small n suffices. The interference bound, by contrast, mixes the W_{1+} , W_1 , W_2 , and `none` arms, including the intermediate Opus baselines (`api_argorder` and trap `none` = 15%), so the four-term Clopper–Pearson sum stays wide at $n = 20$ (upper bounds $-0.08 / -0.11$, not below $-\tau$). Raising Opus to $n = 80$ is the one remaining gap in the consolidated record.

Table 2: Certified incomparability: uniform $(1 - \eta)$ lower bounds $L_{\mathcal{D}}$ in both directions ($\eta = 0.10$). $L_{\mathcal{D}}(W_1, W_2) > 0$ and $L_{\mathcal{D}}(W_2, W_1) > 0$ together certify empirical incomparability. Both bounds are read off the same $2|\mathcal{D}|$ Clopper–Pearson intervals (Proposition 3).

Model	Vendor	n	$L_{\mathcal{D}}(W_1, W_2)$	$L_{\mathcal{D}}(W_2, W_1)$
Haiku-4.5	Anthropic	80	+0.877	+0.831
Sonnet-4.6	Anthropic	40	+0.762	+0.762
Opus-4.8	Anthropic	20	+0.552	+0.552
GPT-5.5	OpenAI	40	+0.762	+0.762
DeepSeek-V4-Pro	DeepSeek	40	+0.347	+0.715
Kimi-K2.6	Moonshot	40	+0.675	+0.762

8.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$, so it prefers W_2 ; yet realized PASS ranks $W_1 \gg W_2$ on all six models (e.g. Haiku 96% vs. 0% at $n = 80$; Sonnet, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%; Opus 100% vs. 10%). On the honest cells relevance *agrees* with PASS, on the API cells relevance ranks $W_1 > W_2$ and so does PASS, and on the encoding cell relevance ranks $W_2 > W_1$ and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The **none** = 0% floor on the trap confirms genuine unguessability and $W_2 = 0\%$ confirms that encoding knowledge is useless there.

8.4.3 Anti-monotonicity: a certified superset that hurts

The superset W_{1+} , which literally *contains* W_1 ’s signal, sharply degrades the API cells relative to W_1 alone. The per-task interference $\Psi_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T) - \text{PASS}(W_2, T) + \text{PASS}(\text{none}, T)$ is certified negative ($\Psi_T \leq -\tau$, $\tau = 0.30$) on at least one cell for five of the six models (Table 3). Opus reproduces the effect with strong point estimates ($\Psi_T = -0.75 / -0.80$) but does not certify at $n = 20$. This is a certified empirical falsification of Blackwell superset-dominance on live models (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it.

The leaky-baseline limitation is a stated protocol. The interference upper bound includes the term $+\text{CP}_{\text{hi}}(p_{\text{none}})$, so it can certify harm only on cells whose no-context baseline floors near zero. This is exactly why anti-monotonicity certifies on **api_argorder** (hardened) for Haiku (**none** dropped from 40% to 1% after hardening, W_1 still 100%) but not on **api_argorder** for GPT-5.5 (**none** = 75%), Kimi (45%), or Opus (15%). We state this as a protocol up front (Remark 6): we certify anti-monotonicity only on cells whose **none** baseline floors near zero, because a leaky baseline mechanically shrinks Ψ_T ’s certifiable magnitude. Table 3 therefore reports, per model, which cell carried the certificate and the collapse on it. We also report the W_1 ceiling alongside every Ψ_T : DeepSeek’s W_1 arm reaches only 75% on **api_post_ok** (an honest capability difference, vs. $\sim 100\%$ for Claude and GPT), so its $\Psi_T = -0.75$ is computed from a lower starting point and is not directly magnitude-comparable to Sonnet’s $100\% \rightarrow 0\%$.

Order control. Reordering the superset does not restore performance. At $n = 80$ on Haiku, **api_post_ok** gives W_{1+} 24% and W_{1+}^{rev} 1% (vs. W_1 98%), and **api_argorder** gives W_{1+} 24% and

Table 3: Anti-monotonicity (interference) certificate on the carrying cell per model. $\Psi_T = \text{PASS}(W_{1+}) - \text{PASS}(W_1) - \text{PASS}(W_2) + \text{PASS}(\text{none})$; “Upper” is the $\eta/4$ Clopper–Pearson upper bound $\bar{\Psi}_T$ of (7); the cell is certified harmful when $\text{Upper} \leq -0.30$. The $W_1 \rightarrow W_{1+}$ column reports the PASS collapse. The certificate clears only on cells whose **none** baseline floors near zero (the leaky-baseline limitation; Remark 6).

Model	Carrying cell	$W_1 \rightarrow W_{1+}$	Ψ_T	Upper	Certified
Haiku-4.5	<code>api_post_ok</code>	98% $\rightarrow$ 24%	−0.74	−0.49	✓
Haiku-4.5	<code>api_argorder</code> (hardened)	100% $\rightarrow$ 1%	−0.97	−0.79	✓
Sonnet-4.6	<code>api_post_ok</code>	100% $\rightarrow$ 0%	−1.00	−0.69	✓
Sonnet-4.6	<code>api_argorder</code>	100% $\rightarrow$ 0%	−1.00	−0.69	✓
GPT-5.5	<code>api_post_ok</code>	100% $\rightarrow$ 0%	−0.95	−0.60	✓
DeepSeek-V4-Pro	<code>api_post_ok</code>	75% $\rightarrow$ 0%	−0.75	−0.36	✓
Kimi-K2.6	<code>api_post_ok</code>	100% $\rightarrow$ 0%	−1.00	−0.69	✓
Opus-4.8	<code>api_post_ok</code>	100% $\rightarrow$ 25%	−0.75	−0.08	, ($n = 20$)
Opus-4.8	<code>api_argorder</code>	100% $\rightarrow$ 5%	−0.80	−0.11	no ($n = 20$)

W_{1+}^{rev} 44% (vs. W_1 100%). On one cell the reversed order is worse and on the other it is better, so order does have a directional effect; but *neither* ordering recovers W_1 alone. The defensible reading is that reordering does not cure the harm, hence the collapse is not a lost-in-the-middle / recency artifact, not that order is irrelevant.

Runtime fingerprint (a descriptive correlate). The harmful W_{1+} arm consistently emits more output tokens than W_1 : +106%/ + 87%/ + 252% across the Haiku collapse and encoding cells (and +12% even on the still-passing encoding superset cell), Sonnet +120%/ + 131%, Opus +204%/ + 204%, GPT-5.5 +281%/ + 274%, DeepSeek +93%/ + 166%, Kimi +203%/ + 249%. We report this as a consistent descriptive correlate, not a mechanism: more output on the failing arm could be effect rather than cause, and the +12% inflation on a cell that still passes undercuts a strict “verbose thrashing causes the collapse” reading.

Robust, but not monotone in capability. The harm is large and robust across the Anthropic capability ladder but is *not* ordered by capability. On the two API cells the $W_1 \rightarrow W_{1+}$ collapse is Haiku 100% $\rightarrow$ 24%/1%, Sonnet 100% $\rightarrow$ 0%/0%, and Opus 100% $\rightarrow$ 25%/5%: the middle model (Sonnet) shows the starkest collapse and the strongest model (Opus) less so. An earlier two-point reading suggested the harm “intensifies with strength”; the third point falsifies that, and we explicitly do not claim a capability trend, only robustness across the ladder.

8.4.4 The model-relative value of private context

The “private” conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The **none**-arm baseline on `api_argorder` forms a clean spectrum: Claude $\sim 0\%$, DeepSeek 2%, Opus 15%, Kimi 45%, GPT-5.5 75% (Table 1). Thus the value of the private context, the model-relative quantity $I(S; W \mid \theta)$ of §3.3, varies continuously across models. It tracks a vendor’s training rather than raw capability: DeepSeek, a different vendor from Anthropic, nonetheless has a tight prior (baselines 0/2/0/0% across `api_post_ok`/`api_argorder`/`enc_amount`/`trap`) like Claude, while GPT-5.5’s prior already holds much of the convention (`api_argorder` 75%, `enc_amount` 55%). Sonnet’s baselines all floor at 0%,

Table 4: Per-cell PASS rates (%) for the carrying cells, by arm. Anthropic rows agentic, others single-shot. “, ” marks an arm not in that run. Hardened `api_argorder` (Haiku) lowered the `none` baseline from 40% to 1% while leaving W_1 at 100%.

Model	Cell	none	W_1	W_2	W_{1+}	W_{1+}^{rev}
Haiku-4.5	<code>api_post_ok</code>	0	98	0	24	1
Haiku-4.5	<code>api_argorder</code> (hard.)	1	100	0	1	44
Haiku-4.5	<code>enc_amount</code>	0	0	100	–	–
Haiku-4.5	<code>trap_store_wire</code>	0	96	0	–	–
Sonnet-4.6	<code>api_post_ok</code>	0	100	0	0	–
Sonnet-4.6	<code>api_argorder</code>	0	100	0	0	–
Sonnet-4.6	<code>trap_store_wire</code>	0	100	0	–	–
Opus-4.8	<code>api_post_ok</code>	0	100	0	25	–
Opus-4.8	<code>api_argorder</code>	15	100	0	5	–
Opus-4.8	<code>trap_store_wire</code>	15	100	10	–	–
GPT-5.5	<code>api_post_ok</code>	5	100	0	0	–
GPT-5.5	<code>api_argorder</code>	75	–	–	–	–
GPT-5.5	<code>enc_amount</code>	55	–	–	–	–
GPT-5.5	<code>trap_store_wire</code>	0	100	0	–	–
DeepSeek-V4-Pro	<code>api_post_ok</code>	0	75	0	0	–
DeepSeek-V4-Pro	<code>api_argorder</code>	2	68	0	–	–
DeepSeek-V4-Pro	<code>trap_store_wire</code>	0	98	0	–	–
Kimi-K2.6	<code>api_post_ok</code>	0	100	0	0	–
Kimi-K2.6	<code>api_argorder</code>	45	–	–	–	–
Kimi-K2.6	<code>trap_store_wire</code>	0	100	0	–	–

so even the stronger Anthropic model does not guess the convention; the value of private context survives the capability ladder. The certified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok` `none` = 5%, `trap` = 0%). We note that this cross-vendor magnitude comparison is the part most exposed to the agentic-vs-single-shot confound (§10): vendor and harness are not separated in the present runs.

Auditable per-cell record. Table 4 gives the per-arm PASS rates for the carrying cells so each certificate can be reconstructed; the full per-cell, per-model record is in the supplementary material and the run logs (`blackwell*_n*.log,json`).

8.5 Generality (a second pair) and harness robustness (single-shot)

Two follow-up experiments address the two largest threats to validity directly.

A second incomparable pair, different domain. We repeated the protocol with a second hand-built pair on a private `cache` module: V_1 , the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False “newly inserted” flag, a `CacheError` type), versus V_2 , a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at $n = 40$ the pair is again certified incomparable ($L_D = +0.774/ +0.496$, both directions clear of zero) with a clean double crossover (V_1 solves the contract and trap cells, V_2 solves key-normalization, all none-baselines floor at 0%), and the dominance control again certifies

one-directionally. Anti-monotonicity reproduces in direction (the superset collapses the contract cell $80\% \rightarrow 28\%$, $\Psi_T = -0.53$) but does not certify at $n = 40$ (the V_1 rescue is only 80% , a smaller gap than pair 1; it would require $n \approx 80$). The relevance trap for this pair was a null, its prompt vocabulary leaned toward V_1 ($\varphi(V_1) = 0.39 > \varphi(V_2) = 0.25$), so relevance *agreed* with PASS, so the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability and dominance results to a new domain*, not a second mis-rank.

Single-shot breaks the harness confound. To separate vendor from harness, we ran an Anthropic model (Haiku) in *single-shot completion* mode, one turn, no tools, code extracted from the returned text, the same regime as the non-Anthropic models. At $n = 40$ the core results survive the non-agentic harness: incomparability is certified ($L_{\mathcal{D}} = +0.762$ both directions), the relevance mis-rank reproduces (trap PASS W_1 90% vs W_2 0%), the dominance control certifies, and anti-monotonicity is certified on `api_argorder` ($\Psi_T = -0.93$, upper bound -0.56). The one regime-dependent quantity is absolute execution reliability: single-shot W_1 rescue on `api_post_ok` is 52% versus 98% agentic (no retries to self-correct), so that cell’s interference does not certify single-shot while its order-theoretic claims do. The order-theoretic findings, certified incomparability, relevance mis-rank, dominance, and at least one certified interference cell, are therefore *not* artifacts of the agentic harness.

9 Discussion

Our results recast a piece of practitioner folklore, “relevance does not transfer across tasks”, as a structural fact about context sources. When two sources are Blackwell-incomparable, the order over them is genuinely partial, and any scalar score, by imposing a total order, must mis-rank them on some task. The empirical program does not merely illustrate this; it instantiates a single hand-built incomparable pair (a private API-call contract W_1 versus a private wire-encoding invariant W_2 on the same `ledger` module) and shows that the resulting double crossover, the relevance mis-rank, and an anti-monotonicity collapse all reproduce across six models and four vendors. We discuss what the evidence does and does not license, and where the formalization is a model rather than a proof of the realized object.

9.1 Theorems are about the idealized decision-maker; the experiments measure a fixed rule

The cleanest way to read the theory is as a two-layer claim, and the two layers must not be conflated. Corollary 1 (universal context dominance) and Theorem 2 (no scalar ranks incomparable sources) are statements about the *Bayes-optimal* user of a source: “ $W_1 \succeq_B W_2$ ” means the optimal achievable value under W_1 dominates that under W_2 for every prior, bounded utility, and decision rule [Blackwell, 1953]. The operational quantity in every experiment, by contrast, is $\text{PASS}(W, T) = \mathbb{E}[v_T(\delta_M(T, w))]$ for a single *fixed, sub-optimal* decision rule δ_M , the language model. “Optimal-user-of- W_2 beats optimal-user-of- W_1 on some decision problem” does not, by itself, imply “the fixed model’s PASS under W_2 exceeds its PASS under W_1 on task T_b .” Lemma 2 supplies the bridge in the direction we need (certified $(M, \mathcal{D})$ -incomparability implies Blackwell incomparability) and isolates the value-attainment hypothesis (Assumption 1) required for the converse. We therefore read the contribution as a δ_M -restricted, $\mathcal{D}$ -restricted incomparability statement, certified directly from PASS data, that is *consistent with* and motivated by the idealized Blackwell incomparability of the underlying coordinates, and, via the bridge, earns the structural conclusion rather than assuming it.

9.2 The estimator is a regret-gap surrogate, not a Le Cam deficiency

Genuine Le Cam deficiency $\delta(\mathcal{E}, \mathcal{F})$ minimizes over garblings and takes a supremum over a class of loss functions [Le Cam, 1986]. Our surrogate $\hat{\delta}_{\mathcal{D}}$ (4) performs *no* garbling minimization, fixes a single sub-optimal rule δ_M , and ranges over a finite task battery $\mathcal{D}$. It is a one-sided decision-restricted *value/regret gap*, not a deficiency in Le Cam’s sense, and we do not prove a two-sided inequality relating it to the true deficiency (for a fixed sub-optimal rule none holds in general in either direction; Remark 4). We accordingly read $\hat{\delta}_{\mathcal{D}}$ as a “decision-restricted value-deficiency surrogate” and demote the Le Cam billing to an analogy that motivates the construction. What is rigorous is the *certificate*: the distribution-free uniform Clopper–Pearson [Clopper and Pearson, 1934] plus Bonferroni [Bonferroni, 1936] lower bound $L_{\mathcal{D}}$ on $\sup_T [\text{PASS}(W_2, T) - \text{PASS}(W_1, T)]$, which we regard as the strongest methodological component and which survives independently of the deficiency framing.

9.3 What the certificate covers, and what it does not

The uniform certificate is the part of the paper we are most confident in. Its correctness rests on paying the multiplicity of the $\sup_T$ selection via a union bound over $2|\mathcal{D}|$ exact intervals rather than absorbing it into a single-sample sup inequality; a naive DKW bound under-covers precisely because the object is a maximum of *differences of two binomial means across distinct cells*, not the sup-deviation of one empirical CDF from its own (Remark 5). Two qualifications follow. First, the effective support carrying the certificate is small: although $\mathcal{D}$ has four cells, $L_{\mathcal{D}}(W_1, W_2)$ is realized on one encoding cell and $L_{\mathcal{D}}(W_2, W_1)$ on one API cell, so the certified incomparability is a two-cell crossover replicated across models, not a sample from a task distribution. We therefore describe $\mathcal{D}$ as a fixed adversarial battery rather than a “task distribution.” Second, each certified claim per model is controlled *marginally*, not jointly: incomparability (at $\alpha = \eta/(2|\mathcal{D}|)$ over $2|\mathcal{D}|$ intervals), the dominance control, and the interference certificate (at $\alpha = \eta/4$) draw from the same run at the same $\eta = 0.10$, so the family-wise error across all certified claims on a model is not simultaneously controlled (§6.5). Readers should treat each marginal certificate as valid in isolation; a strict joint statement would require a wider global Bonferroni budget.

9.4 Anti-monotonicity is real, certified, but not capability-ordered

The interference object $\Psi_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T) - \text{PASS}(W_2, T) + \text{PASS}(\text{none}, T)$ measures whether appending the (irrelevant-to-this-task) W_2 to a sufficient W_1 harms performance. We certify $\Psi_T \leq -0.30$ on at least one cell for five of six models (Table 3), with collapses as stark as $100\% \rightarrow 0\%$ ($\Psi_T = -1.00$, upper bound -0.69) on Sonnet and Kimi. This is a certified empirical falsification of superset-dominance for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone-submodular scalar context-value score can represent.

Two corrections follow from the data. **(i) The harm is not monotone in capability.** An early two-point read (Haiku then Sonnet) suggested the collapse “intensifies with model strength”; adding the third Anthropic model (Opus) falsified this. On `api_post_ok` the collapse is $100 \rightarrow 24\%$ (Haiku), $100 \rightarrow 0\%$ (Sonnet), $100 \rightarrow 25\%$ (Opus), and on `api_argorder` $100 \rightarrow 1\%$, $100 \rightarrow 0\%$, $100 \rightarrow 5\%$ respectively, the middle model (Sonnet) is the starkest and the strongest (Opus) is less so. The defensible claim is robustness across the capability ladder, *not* a capability trend. **(ii) The interference certificate is structurally cell-selected.** Its upper bound includes $+\text{CP}_{\text{hi}}(p_{\text{none}})$, so it can only certify harm on cells whose none-baseline floors near zero. This is exactly the set of cells that are unguessable for the model under test, which couples the certifiable-harm set to

the model-relative value of private context (§9.5). We treat this as a protocol (Remark 6): we certify anti-monotonicity only on floor-baseline cells, because a leaky baseline mechanically shrinks the certifiable Ψ_T . On `api_argorder` the harm is certified for Haiku (hardened baseline 1%) and Sonnet (0%) but not for GPT-5.5 (baseline 75%), DeepSeek, Kimi (45%), or Opus (15%).

9.5 The value of private context is model-relative and tracks vendor training

The value of a private convention, $I(S; W \mid \theta)$, is itself a model-relative (θ -dependent) object, and the data give a clean continuous map of it. The none-arm baseline on `api_argorder`, how often a model reproduces the private convention with no context supplied, forms a spectrum: Claude $\sim 0\%$, DeepSeek 2%, Opus 15%, Kimi 45%, GPT-5.5 75%. We do not claim this tracks capability. The decisive piece of evidence is DeepSeek: it has a tight, near-zero prior like Claude despite being a different vendor, while GPT-5.5’s prior is wide (55%–75% guessing on `api_argorder/enc_amount`). A “stronger-model-knows-more” account predicts the spectrum to follow capability; it does not. We read $I(S; W \mid \theta)$ as tracking a vendor’s training distribution rather than raw capability, which makes the theory’s θ -dependence concrete and caps which cells certify per model.

9.6 Mechanism: the runtime fingerprint is a correlate, not an explanation

Every collapse is accompanied by output-token inflation on the harmful arm: W_{1+} emits between +87% and +252% more output tokens than W_1 on the API collapse cells (e.g. Haiku +106%/ +87%/ +252%; Sonnet +120%/ +131%; GPT-5.5 +281%/ +274%). One might read this as “verbose thrashing” causing the failure, but the relationship is correlational and uncontrolled: the same superset arm also inflates output by +12% on the *still-passing* encoding cell, which undercuts a clean “verbosity causes collapse” account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

10 Limitations and threats to validity

We collect the scope restrictions here.

Scaling the selector is the central open problem. PROBE-SELECT (§7) is curation-scale: building the certified order on m candidate sources costs $m \cdot |\mathcal{D}| \cdot n$ verifier-gated model calls, which is practical for choosing among a handful of private conventions or curating a few-shot pool, but not for retrieval over a corpus of 10^6 passages. We prove the selector’s properties and validate the estimator that powers it (§8), but we do *not* benchmark a corpus-scale system against strong dense rerankers. Closing this gap, most plausibly by amortizing the probe into a learned deficiency predictor, or by using PROBE-SELECT as a partial-order re-ranking head over a cheap scalar shortlist, is, in our view, the most important direction this work opens, and the one on which its practical impact depends.

Source-pair generality (partially addressed). The cross-model record (six models) rests on one adversarially hand-built pair, the API contract W_1 versus the wire-encoding invariant W_2 . To test whether the partial order is a *pattern* rather than a one-off, we ran a second pair in a different domain, a `cache-module` call contract V_1 versus a key-normalization invariant V_2 , which is again certified incomparable on Haiku ($L_{\mathcal{D}} = +0.774 / +0.496$) with a certified one-directional dominance control (§8.5). Incomparability and dominance therefore hold for *two* pairs in *two* domains, not one. The remaining content gaps are narrower: pair 2’s anti-monotonicity is only directional at $n = 40$ (would certify at $n \approx 80$), pair 2’s relevance trap was a null (its prompt was not lexically V_2 -leaning),

and both effects on pair 2 are shown for a single model (Haiku), so the *cross-vendor* record is still single-pair. A third domain and a vendor-general strong mis-rank remain inexpensive next steps.

Vendor is perfectly confounded with harness. The Anthropic models (Haiku, Sonnet, Opus) run agenticly via `claude -p` with up to six turns and intermediate verifier feedback; the non-Anthropic models (GPT-5.5, DeepSeek, Kimi) run as single-shot completion followed by code extraction over an OpenAI-style API. Every Anthropic row is therefore agentic and every non-Anthropic row single-shot, so “cross-vendor” and “cross-harness” cannot be separated in the six-model dataset. The binary per-model claims (incomparability certified; relevance mis-rank; at least one certified anti-monotonicity cell) are within-model and do not depend on cross-arm comparability. To break the confound for the order-theoretic claims, we ran one Anthropic model (Haiku) in the single-shot regime (§8.5): incomparability, the relevance mis-rank, dominance, and one certified interference cell all survive the non-agentic harness, so these findings are not artifacts of the agentic arm. The cross-model *magnitude* comparisons, the 0/2/15/45/75% value spectrum and the Ψ_T severity ladder, still depend on cross-arm comparability (the harness-crossed run was on one model, and absolute execution reliability *is* regime-dependent: W_1 rescue 98% agentic vs. 52% single-shot on `api_post_ok`), so those magnitude comparisons should still be read as suggestive. We argue single-shot is arguably the *cleaner* probe of whether a model uses the supplied context (no agentic-exploration confound).

Decision-restricted, $\mathcal{D}$ -restricted scope. $\hat{\delta}_{\mathcal{D}}$ certifies dominance and incomparability *for a given model on a fixed task battery*, not the full universal Blackwell order. This restriction is exactly what makes the object computable from PASS data, and within $\mathcal{D}$ it remains a decision-independent-of-task guarantee, but it is not the universal order, and we do not claim it is.

Sampling regime and per-draw independence. The certificate assumes n i.i.d. Bernoulli PASS outcomes per cell (Assumption 2). We report the exact decoding parameters, seeding policy, and turn budget in §12, and we note the assumption is most delicate for the agentic arm, where each draw is a multi-turn trajectory; we treat each completion as a separate draw and depend on per-draw independence for the binomial intervals. The asymmetry by which $n = 20$ certifies incomparability for Opus but not its interference is principled rather than ad hoc (§6.6): stark 0/1 cells make the Clopper–Pearson endpoints collapse, so few draws suffice, whereas the interference contrast mixes intermediate baselines and needs more, so the two certificates have different effective- n requirements by design.

Leaky baselines bound the certifiable set. As in §9.4, anti-monotonicity can only be certified on cells whose none-baseline floors near zero. We treat “certify only on floor-baseline cells” as a stated protocol (Remark 6), and report per model which cell carried the certificate and its baseline (Table 1, Table 3).

Relevance is demonstrated only for lexical cosine. The mis-rank ($\varphi(W_1) = 0.36 < \varphi(W_2) = 0.44$ on `trap_store_wire`, while realized PASS is $W_1 \gg W_2$, e.g. 96–100% vs 0%) is shown for bag-of-words cosine relevance. The theorem rules out any *query-independent* scalar (Theorem 2); the trap is an *empirical* patch against *query-conditioned* relevance (Proposition 2) and is shown for one concrete scalar. A learned dense reranker might or might not mis-rank the trap; reproducing the mis-rank with an off-the-shelf embedding reranker would strengthen the strong-form claim, which we currently restrict to lexical relevance.

Marginal, not joint, certification. The incomparability, dominance, and interference certificates drawn from one run are each controlled at $\eta = 0.10$ in isolation; we do not claim joint family-wise control across all certified claims per model (§6.5).

Synergy (positive interference) is untested. By construction W_1 and W_2 are task-disjoint, so the experiments measure negative interference only; a complementary pair (two facts a task needs *jointly*) would test whether interference is sign-indefinite (i.e. whether a superset can also *help* beyond its parts). We do not measure this and make no claim about positive synergy.

11 Conclusion

Context selection for language models is, structurally, a partial-order problem rather than a scalar one. When two context sources are Blackwell-incomparable, no scalar score can rank them correctly for all tasks (Theorem 2), which gives a formal account of the field’s “relevance does not transfer” folklore. We made the order measurable with a decision-restricted value-deficiency surrogate $\hat{\delta}_{\mathcal{D}}$ equipped with a distribution-free uniform Clopper–Pearson/Bonferroni certificate, and we instantiated the theory on six models across four vendors. On every model we certify ($\geq 90\%$) that two private context sources are empirically incomparable; on every model a real query-conditioned lexical relevance score mis-ranks them; and on five of six models we certify a superset of a sufficient source sharply degrading PASS (a collapse as deep as $100\% \rightarrow 0\%$), a certified falsification of superset-monotonicity for these fixed rules. The value of private context $I(S; W \mid \theta)$ is itself model-relative, forming a continuous 0%–75% spectrum that tracks vendor training rather than raw capability.

On scope: the theorems concern the idealized Bayes-optimal user of a source; the realized crossover, anti-monotonicity, and value-of-private-context are properties of *fixed* models measured directly, bridged to the structural claim by Lemma 2. The estimator is a regret-gap surrogate, not a Le Cam deficiency; the certificate, which we regard as the most robust contribution, is correct and survives that re-labeling. The evidence now spans two hand-built source pairs in two domains and a single-shot replication that preserves the order-theoretic claims; the cross-vendor record remains single-pair, vendor is confounded with harness for the magnitude comparisons, and synergy is untested. The practical reading is modest and actionable: selection should be treated as partial-order-aware rather than scalar top- k , the right operational object is a certified deficiency surrogate rather than a relevance scalar, and “the best context” should be expected to be model-relative and only partially ordered. The remaining strengthenings, a vendor-general strong mis-rank, an embedding-reranker relevance baseline, a third domain, and certifying pair-2 anti-monotonicity at $n \approx 80$, are the natural next steps, and the experimental record indicates each is cheap to run.

12 Reproducibility

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The harness (`tokenbench/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a `stdlib` regularized incomplete beta), is cross-OS, and implements the surrogate $\hat{\delta}_{\mathcal{D}}$, the uniform incomparability certificate (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference certificate, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic CLI (`claude -p`), the Azure Responses API, and OpenAI-compatible chat

completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

Protocol. The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type); return code 0 is PASS. The four certified cells are `api_post_ok`, `api_argorder` (hardened so `memo` arrives via a parameter, forcing the keyword-only fact and dropping its baseline from 40% to 1%), `enc_amount`, and `trap_store_wire`; arms are $\{\text{none}, W_1, W_2, W_{1+}\}$ plus $W_{1+}^{\text{rev}} = W_1 ++ W_2$ for the order control. PASS is measured over n i.i.d. completions per cell and the certificate uses $\eta = 0.10$. The Anthropic arms run agentially (`claude -p`, up to six turns); the non-Anthropic arms run single-shot (one completion + code extraction). We note for transparency that this design choice, agentic for Anthropic, single-shot for the others, is the vendor/harness confound discussed in §10.

Artifacts. Per-model run logs and result JSONs are in the repository under the pattern `blackwell*_n*.log,json`, including `blackwell_n80_run.log,json` (the $n = 80$ Haiku bundled certification, 1600 calls), `blackwell_argorder_n80.json,log` (the hardened second anti-monotonicity cell), `blackwell_sonnet_n40`, `blackwell_opus_n20`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, and `blackwell_kimi_n40`. The full experimental record with exact per-cell figures is in `EXPERIMENT-BLACKWELL.md` (§§10–13) and the theory hardening (formalization and proof sketches) in `BLACKWELL.md` (§10).

Compute and cost. Total measured spend was approximately \$170 on Anthropic (`claude -p`): roughly \$123 across the Haiku and Sonnet anchors plus the Opus run (\$49.84) and the order-control and second-cell runs. The non-Anthropic runs (GPT-5.5, DeepSeek, Kimi) used Azure quota; the Azure Responses API returns no cost field, so those rows report a \$0 cost artifact while token counts are captured, the \$170 figure is the Anthropic spend, not the total cost of all six models. The two highest-value strengthenings flagged in the record (a second source pair on Haiku, and one harness-crossed run) are each estimated in the tens of dollars.

Reference verification (author note). *The bibliography was assembled in part with automated literature search, which has produced incorrect arXiv identifiers and claims in this project before, so every entry was subsequently web-verified against arXiv/DBLP/publisher records. The outcome: zero fabricated references, all entries resolve to real papers, but eight had incorrect fields (now corrected; e.g. the BED-LLM identifier was repointed from 2604.00414, a different paper, to 2508.21184), and four in-text claims were re-sourced or softened where the cited paper did not support the sentence (most importantly, the “relevance does not transfer across tasks” folklore is now stated without attaching it to a paper that is not about it). A final line-by-line check against the primary sources is still recommended before submission; the classics (Blackwell 1953; Le Cam 1964/1986; Ethayarajh et al. 2022) are author-verified.*

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